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CLIP it real

MeetUp Grenoble

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Foundation models : a « paradigm shift » ¹ in AI ?

On the Opportunities and Risks of Foundation Models

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[1] *On the Opportunities and Risks of Foundation Models*, Bommasani (Stanford) (2022)

Foundation models : a « paradigm shift » ¹ in AI ?

How do we teach a computer to detect if an image contains an apple ?



1990s

2010s

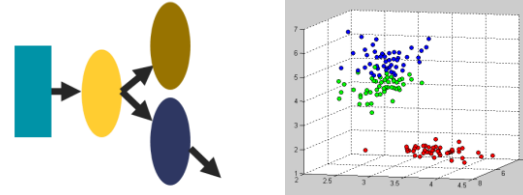
2020s

Specialized algorithms



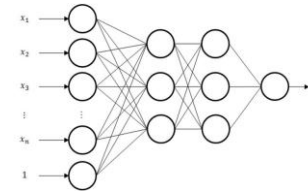
- **Dataset** : ~ 100 images (for human training).
- **Instruction** : « If you find a number of red pixels over this threshold and a shape round enough, image contains an apple. »

«Traditional » (GOF) Machine Learning



- **Dataset** : ~ 1K labeled images with and without apples.
- **Instruction** : « Look for red and circles to figure out if image is apple.»
- **Homogenization of** : classification algorithms.

Deep Learning



- **Dataset** : ~ 1M labeled images with and without apples.
- **Instruction** : « Figure out if this image contains an apple.»
- **Homogenization of** : feature extraction algorithms.

Foundation models



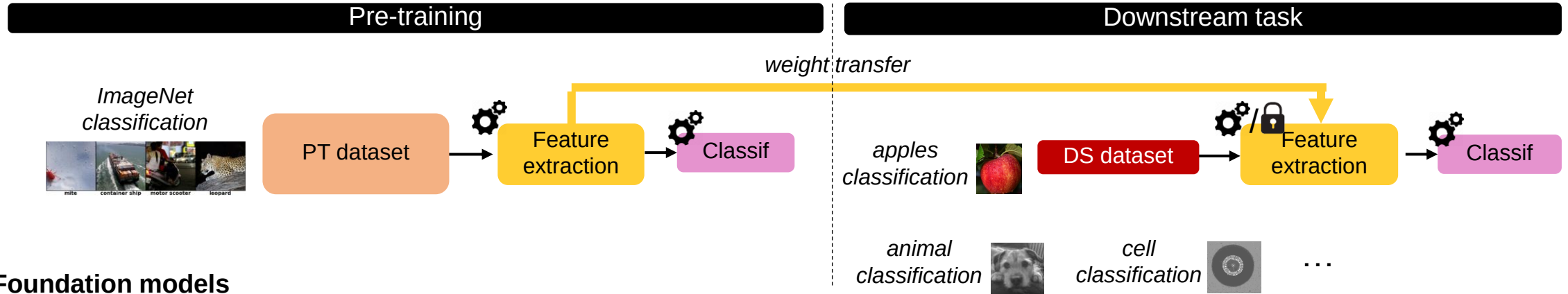
- **Dataset** : ~ 1B weakly labeled/unlabeled images not nec. apple related.
- **Instruction** : « Figure out if this image contains an apple.»
- **Homogenization of** : tasks.

[1] *On the Opportunities and Risks of Foundation Models*, Bommasani (Stanford) (2022) (figure is also inspired by that paper)

Foundation models scale up pre-training

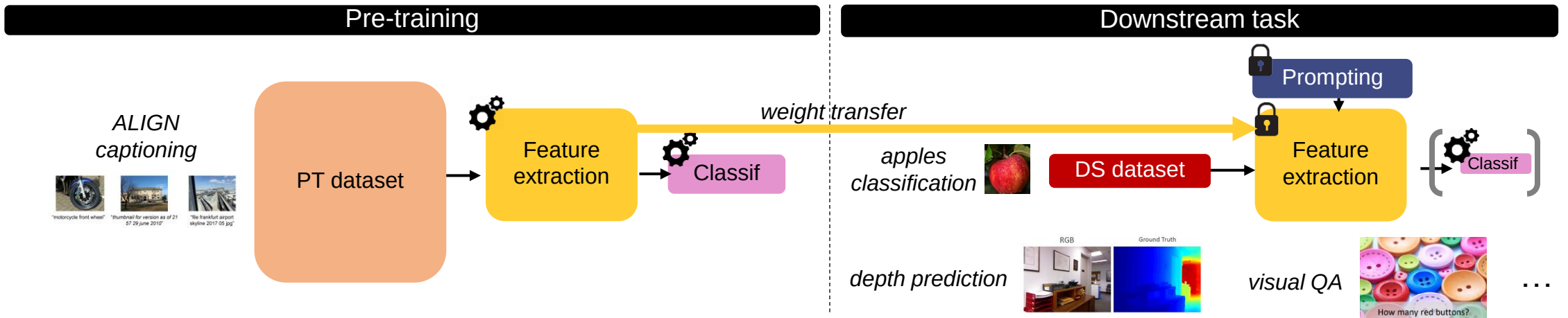
- **Traditional Deep Learning**

- Pre-training (PT) : large dataset, generic task.
- PT and downstream (DS) tasks are often « close ».
- Fine-tuning on DS task.

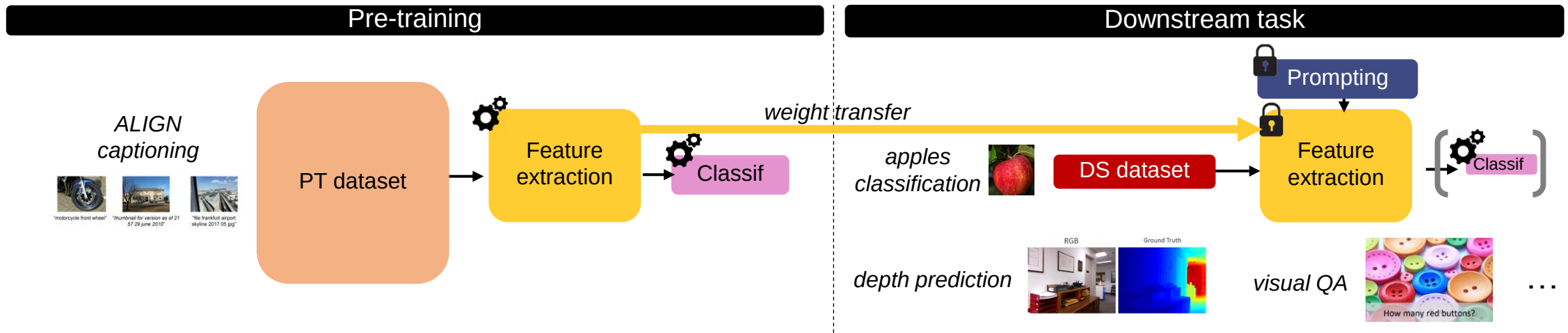


- **Foundation models**

- PT : **very** large dataset, **very** generic task.
- PT and DS tasks can be very different.
- Light / no fine-tuning on DS task.



Foundation models scale up pre-training



↘↘ fine-tuning

↗↗ range of DS tasks

1

A foundation model is any model that is **trained on broad data** (...) that can be **adapted** (e.g., fine-tuned) to a **wide range of downstream tasks** (...). From a technological point of view, [they] are not new — they are based on deep NNs and [pre-training], which have existed for decades. However, the sheer **scale and scope** of foundation models from the last few years have stretched our imagination of what is possible.

1

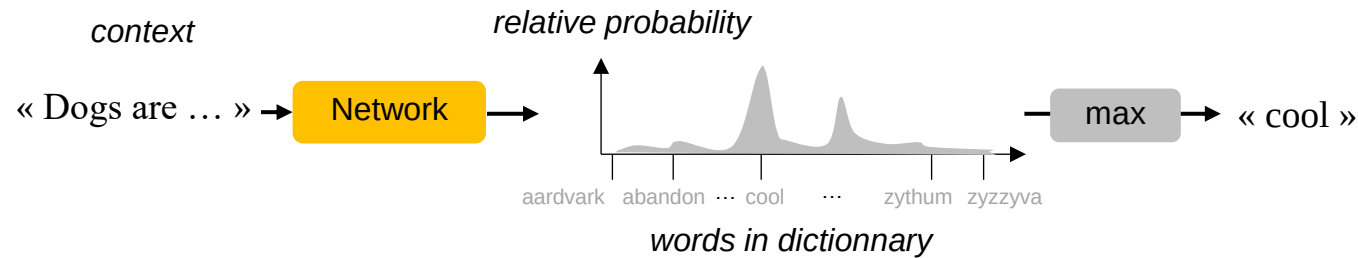
Transfer learning is what makes foundation models possible, but scale is what makes them powerful. Scale required three ingredients:

- (i) improvements in **computer hardware**, e.g., GPU throughput and memory have increased 10× over the last four years,
- (ii) the development of the **Transformer** (...) that leverages the parallelism of the hardware to train much more expressive models than before,
- (iii) the availability of **much more training data**.

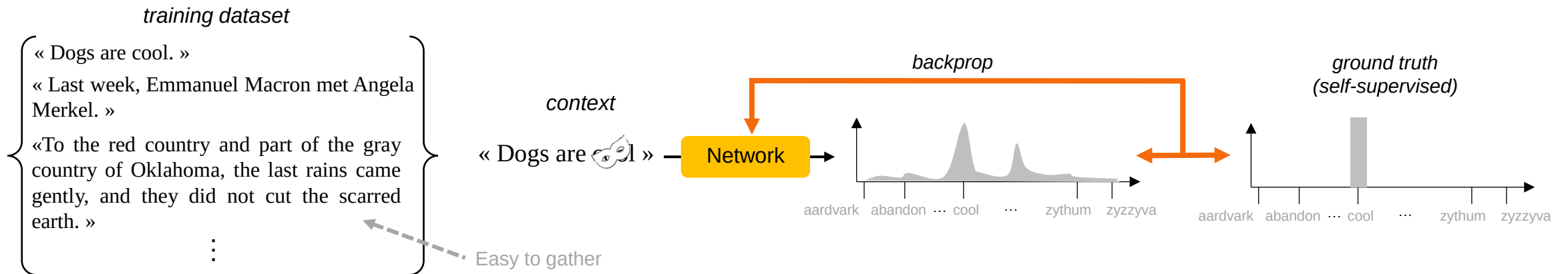
Foundation models started in NLP

→ Straightforward generic PT task :

- ❖ **Language modeling** : understand language,
i.e., be able to **predict probability distribution of unknown words given a context**.



→ Easy to gather very large datasets : **self-supervised learning**.





What of vision ?

The curious case of vision datasets : while GPU (...) power and model sizes have increase[d] (...), **size of the largest training dataset has surprisingly remained constant.**

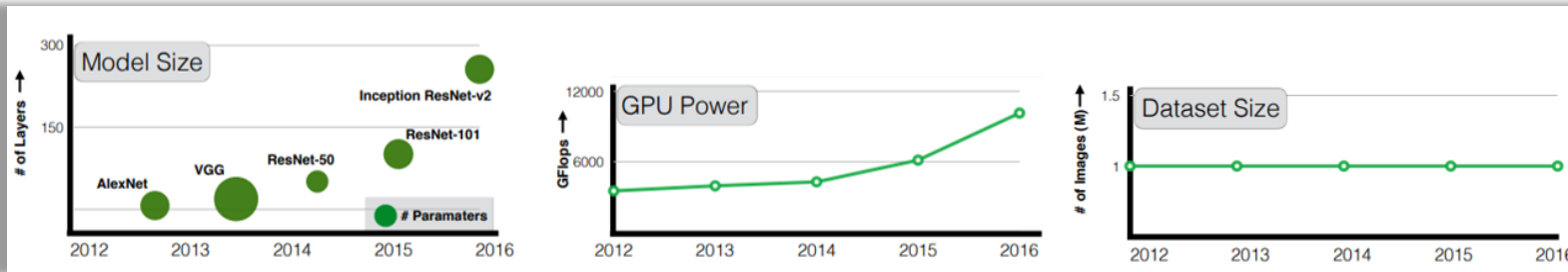
1

❖ ImageNet²



container ship motor scooter leopard

images : 1M
source : internet
labeling : manual



1

[1] Revisiting Unreasonable Effectiveness of Data in DL Era, Sun (2017)

[2] ImageNet: A large-scale hierarchical image database, Deng (2009)

How to scale CV datasets ?

Self-supervise in image space



e.g. in-painting²



(a) Input context

(d) Context Encoder
(L2 + Adversarial loss)

Automatic labeling in « ML format »



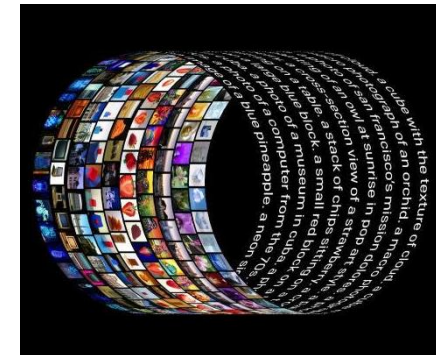
e.g. Instagram-3,5B³



Automatic labeling with natural language supervision



e.g. CLIP⁴



[1] *On the Opportunities and Risks of Foundation Models*, Bommasani (Stanford) (2022)

[2] *Context encoders : feature learning by inpainting*, Pathak (2016)

[3] *Exploring the Limits of Weakly Supervised Pretraining*, Mahajan (2018)

[4] *Learning Transferable Visual Models From Natural Language Supervision*, Radford (OpenAI) (2021)

NLS is easily scalable

Most images on the web are associated with a natural language text rather than a « ML-format » label.

→ Easy to scale NLS datasets by simple web crawl.



dog



my dog Rex :
he's so cute !

❖ Yahoo Flickr dataset¹



images : 90M
source : Flickr
labeling : uploader captions

❖ WebImageText²



images : 400M (~ GPT-2)
source : internet
labeling : « associated » text

❖ ALIGN dataset³



"file london barge race 2 jpg"

images : 1.8B
source : internet
labeling : alt-text

Noisy...
but big is better than clean.



The **scale** of our corpus can **make up for its noise** and leads to SotA representations.



student housing by
lungaard tranberg
architects in copenhagen
click here to see where
this photo was taken

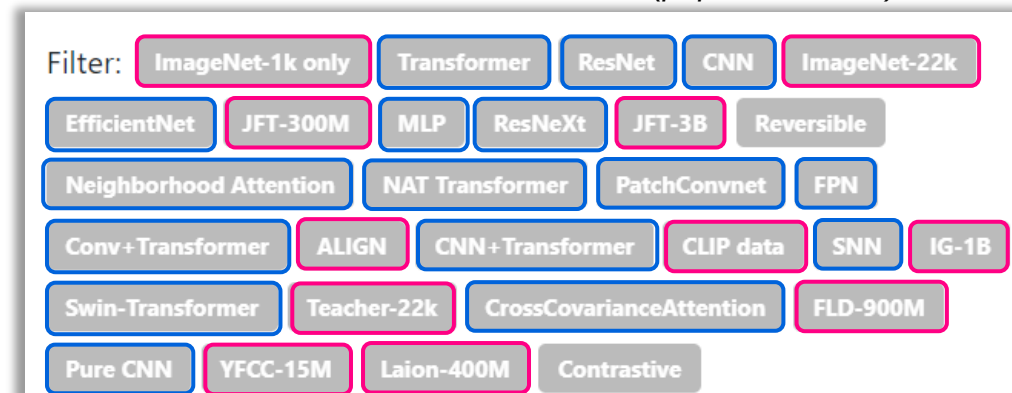


article in the local
paper about all the
unusual things found
at otto's home



this was another one with my old digital
camera i like the way it looks for some things
though slow and lower resolution than new
cameras another problem is that it's a bit of
a brick to carry and is a pain unless you're
carrying a bag with some room it's nearly x x
and weighs ounces new one is x x and weighs
ounces i underexposed this one a bit did
exposure bracketing script underexposure on
that camera looks melty yummy
gold kodak film like

SotA : weight shift from **architecture** to **dataset**
filters on IN classification leaderboard (paperswithcode)



[1] YFCC100M: The New Data in Multimedia Research, Thomee (2016)

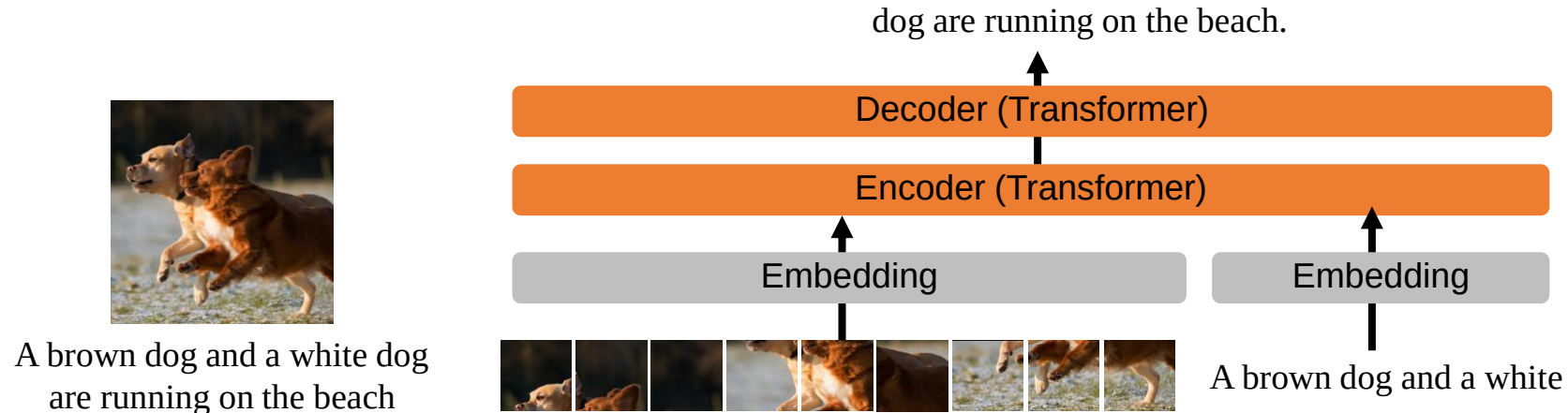
[2] Learning Transferable Visual Models From Natural Language Supervision, Radford (OpenAI) (2021)

[3] Scaling Up Visual and Vision-Language Representation Learning (...), Jia (Google) (2021)

Self-supervision with NLS : generative methods

Learn a joint visual and text representation through masking.

❖ Simple Vision-Language Model (SimVLM)¹



- « Prefix LM » : add image as prefix to LM task → many other fusion implementations are possible²

[1] *SimVLM : Simple visual language model pretraining with weak supervision*, Wang (Google) (2022)

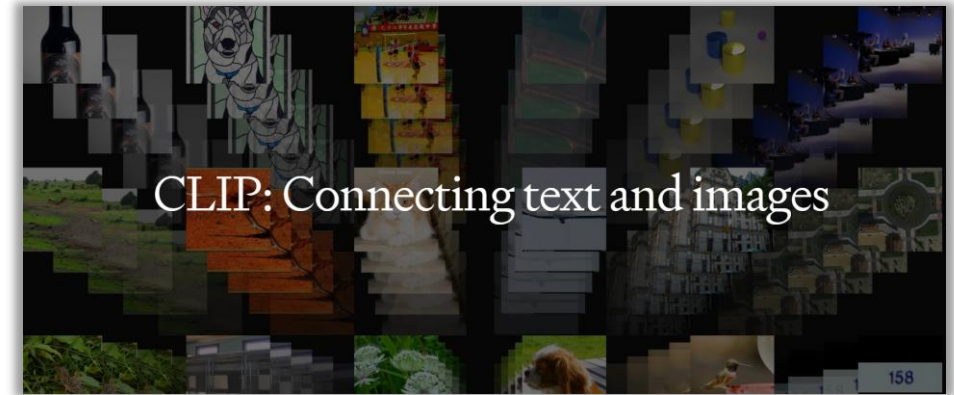
[2] *FLAVA: A Foundational Language And Vision Alignment Model*, Singh (Meta) (2022)

[3] *Learning Transferable Visual Models From Natural Language Supervision*, Radford (OpenAI) (2021)

Self-supervision with NLS : contrastive methods

❖ Contrastive Language-Image Pre-training (CLIP)¹

- FM using NLS and **contrastive learning**
- 3rd most cited AI paper in 2021 (after AlphaFold² and Swin³).
- Building block of further widely-used FMs.



4



5

[1] *Learning Transferable Visual Models From Natural Language Supervision*, Radford (OpenAI) (2021)

[2] *Highly accurate protein structure prediction with AlphaFold*, Jumper (DeepMind) (2021)

[3] *Swin Transformer: Hierarchical Vision Transformer using Shifted Windows*, Liu (Microsoft) (2021)

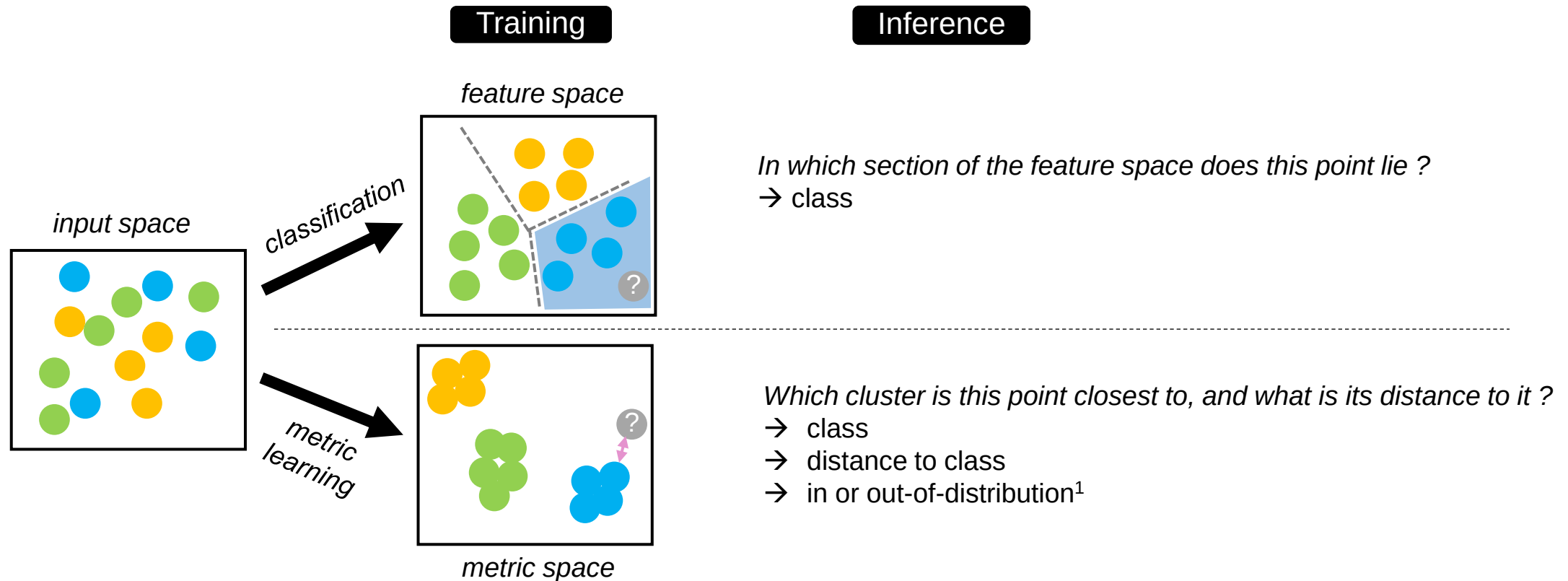
[4] *Segment Anything*, Kirillov (Meta) (2023)

[5] *Hierarchical Text-Conditional Image Generation with CLIP latents*, Ramesh (OpenAI) (2021)

Contrastive loss

Born in the field of metric learning.

❖ **Metric Learning** : compute feature space where distance is correlated to some semantic distance in input space.



Applications : many classes, few points per class, important to know if in/out distribution, e.g. face recognition².

[1] Out-of-distribution detection based on distance metric learning, Yang (2020)

[2] FaceNet: A Unified Embedding for Face Recognition and Clustering, Schroff (2015)

Contrastive loss

Classification loss has no incentive to create « metric space » (« only » focused on class partitionning)

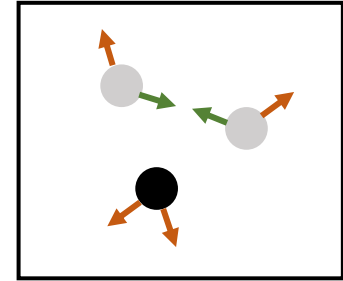
→ need new « contrastive » loss¹.

two points in input space

$$L(x_i, x_j, \theta) =$$

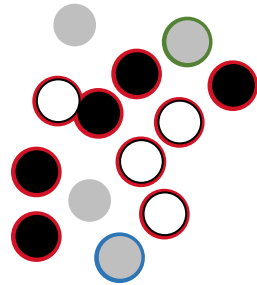
parameters of network f

→ minimal when « same » points are projected close and « different » ones far away



metric space

Efficient implementation : multi-class N-pair loss^{2,3} (a.k.a. InfoNCE loss, NT-Xent...)



batch

$$L(x, x_+, \{x_-\}, \theta) =$$

→ minimal when paired points have similar projection ($f(x)f(x_+)$ high)
and unpaired points have different projection ($f(x)f(x_-)$ low)

[1] Learning a similarity metric discriminatively, with application to face verification, Shopra, LeCun (2005)

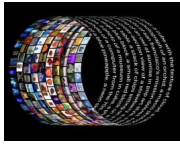
[2] Improved Deep Metric Learning with Multi-class N-pair Loss Objective, Sohn (2016)

[3] Representation Learning with Contrastive Predictive Coding, van dan Oord (Google) (2019)

CLIP : training

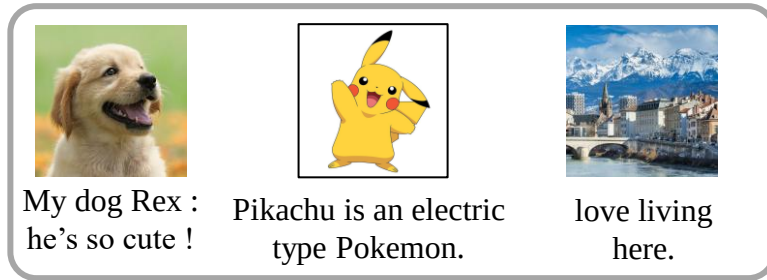
Dataset :

WebImageText



images : 400M (~ GPT-2)
source : internet
labeling : « associated » text

Training batch :

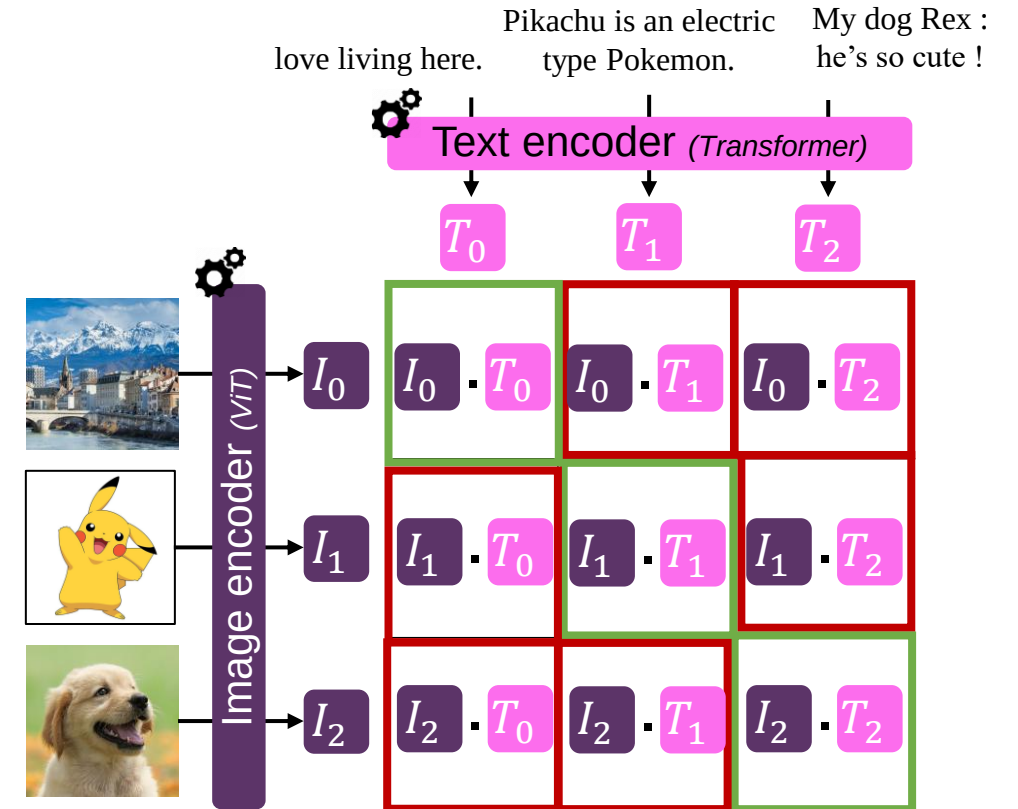


Loss :

$$-\frac{1}{2} \ln \left(\frac{e^{I_i T_i}}{\sum_i e^{I_i T_i} + \sum_j e^{I_i T_j}} \right)$$

correct image-text pair for image i

all other incorrect image-text pairs for image i

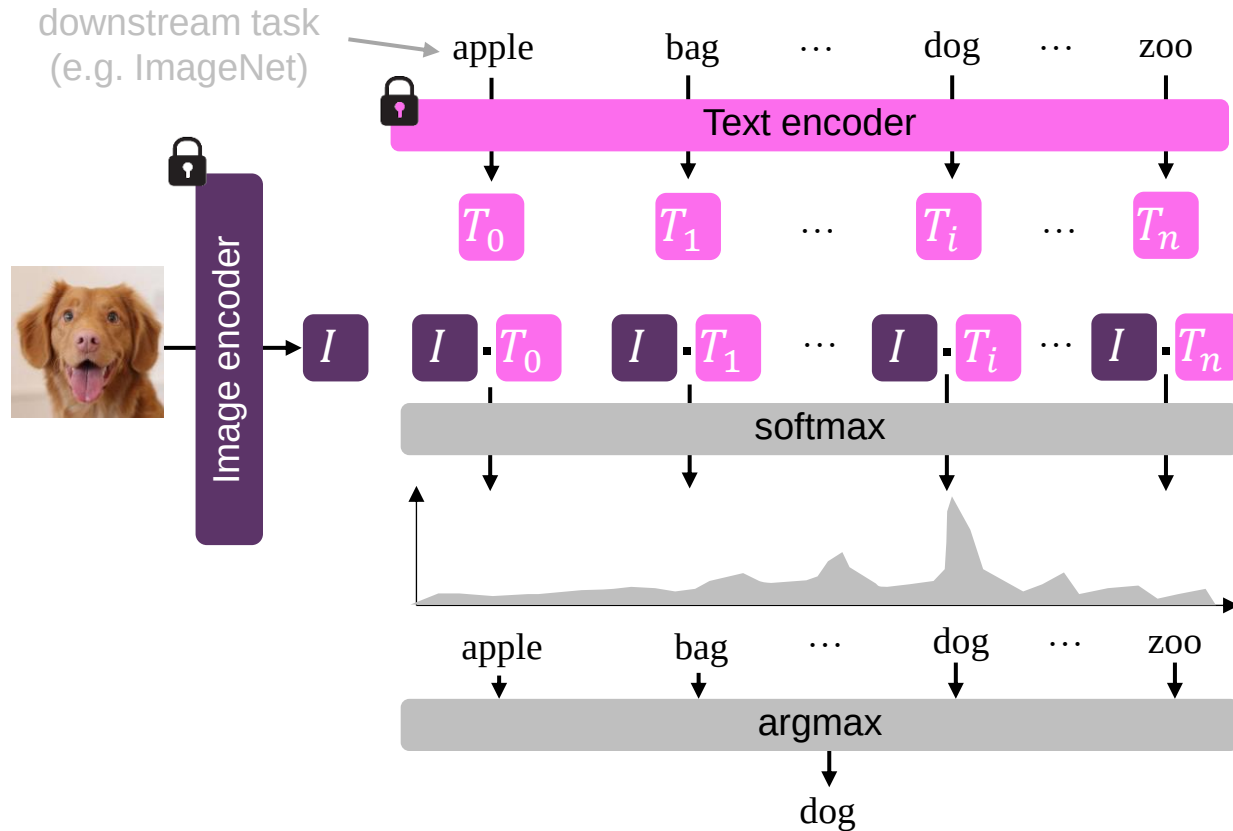


Our initial approach (...) jointly trained an image CNN and text transformer (...) to predict the caption of an image. However, we encountered difficulties (...) : Try[ing] to **predict the exact words** of the text accompanying each image [is] a **difficult task due to the wide variety of descriptions** (...) that co-occur with images.

Recent work (...) has found that **contrastive objectives can learn better representations than their equivalent [generative] objective**.

CLIP : inference

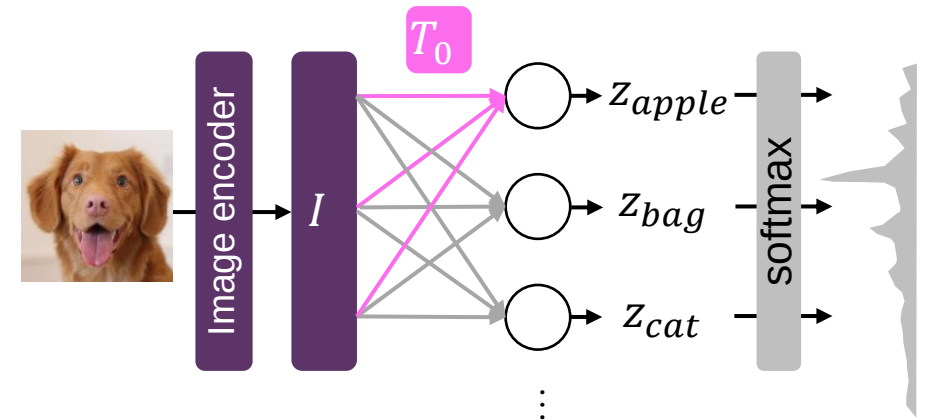
Dataset size big enough to enable **zero-shot** (i.e. no fine-tuning) inference (→ LLMs)



Performance on IN top-1 : 76%, ~ supervised ResNet-50

Prompt engineering¹ : ~ +5 points. « apple » → « a picture of an apple »

When interpreted this way, the **image encoder** is the computer vision backbone which computes a feature representation for the image and the **text encoder** (...) generates the weights of a linear classifier.

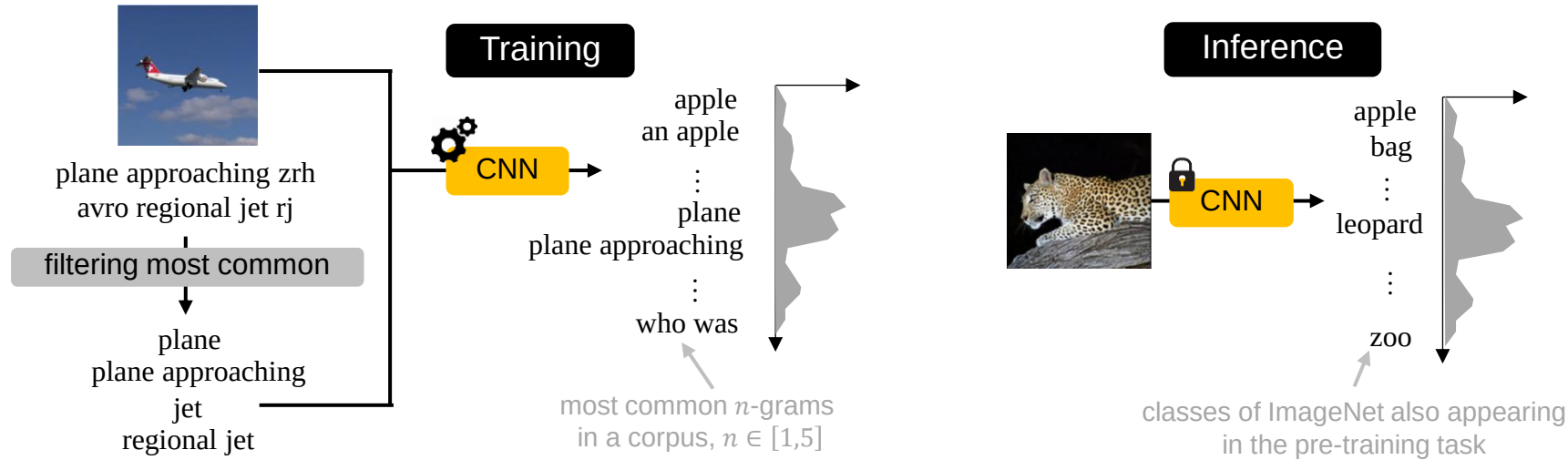


CLIP can be applied to **any visual classification benchmark** by simply providing the names of the visual categories to be recognized, similar to the “zero-shot” capabilities of GPT-2 and GPT-3.

[1] Making pre-trained language models better few-shot learners, Gao (2020)

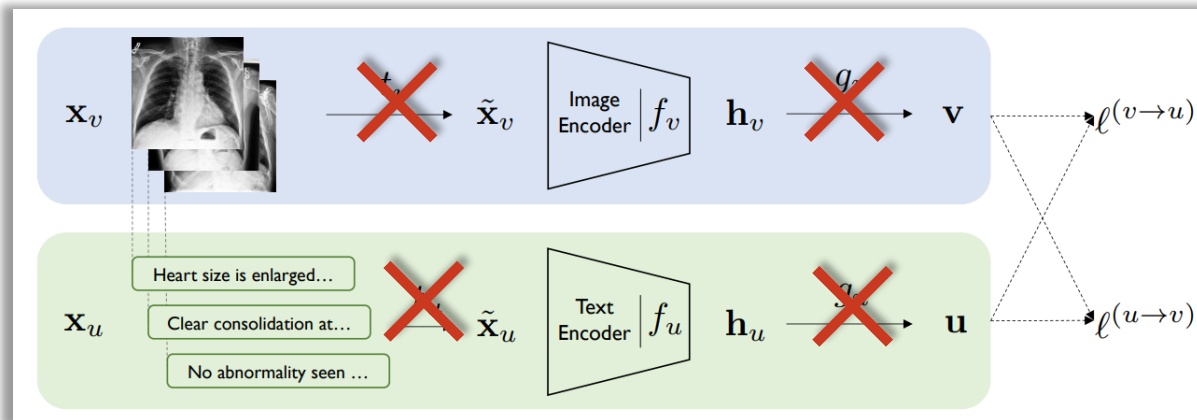
CLIP : Inspirations

❖ **Visual n -grams¹** : large pre-training with NLS + ZS on downstream tasks.



- Dataset : YFCC
- Zero-shot to IN : 11% accuracy

❖ **Contrastive Visual Representation learning from Text (ConVIRT)²** : contrastive alignment of images and text.



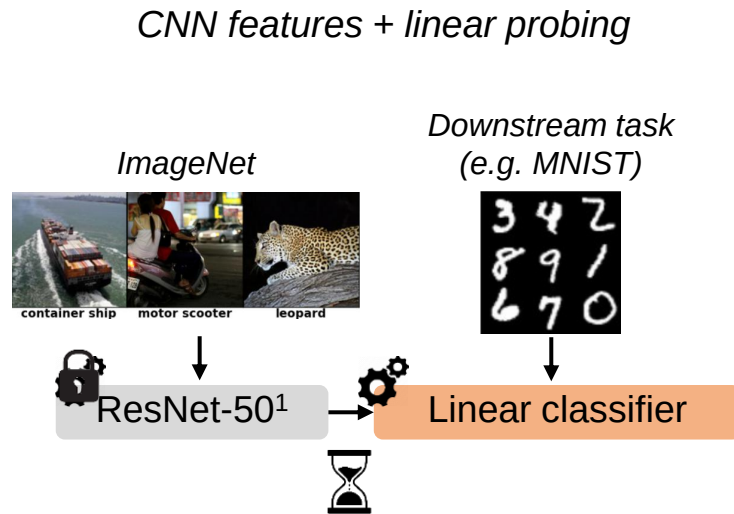
- Dataset : 200k image-text pairs
- Main difference in CLIP : scale and scope
- CLIP training is simplified

[1] *Learning Visual N-Grams from Web Data*, Li (2016)

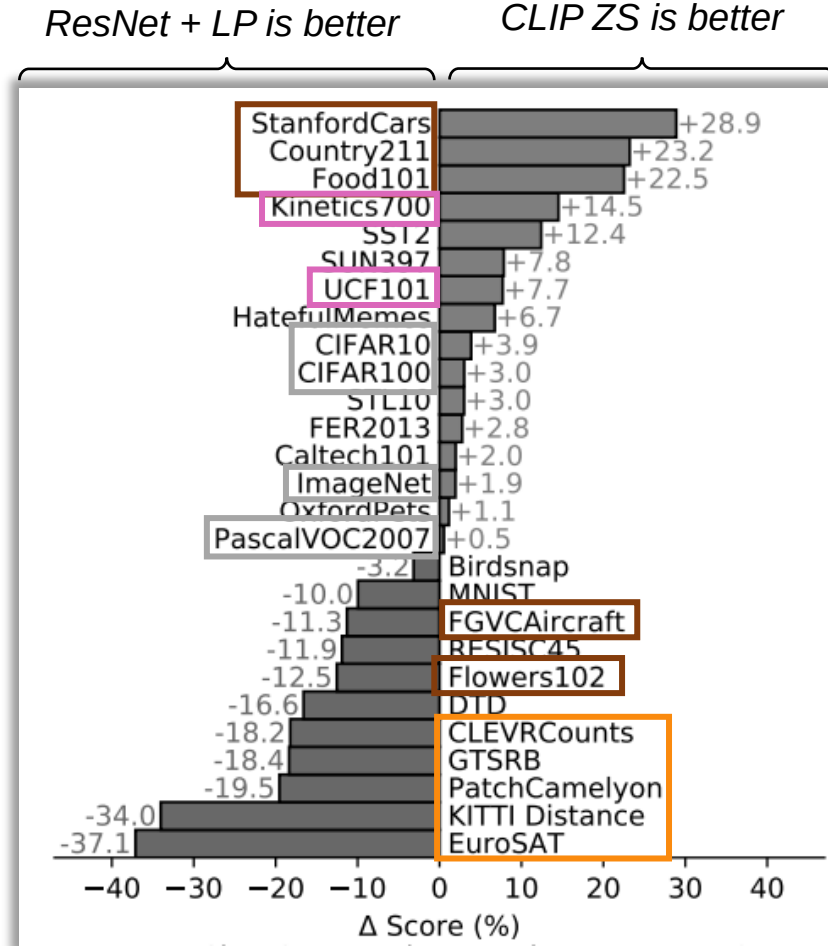
[2] *Contrastive Learning of Medical Visual Representations from Paired Images and Text*, Zhang (2020)

CLIP : Performance (zero-shot)

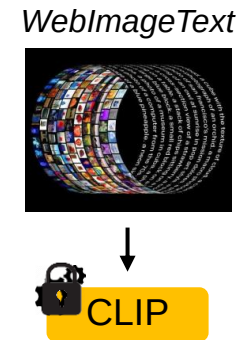
Comparison on several downstream tasks



- General classification : similar performance.
- Fine-grained classification : depends on relative amount of labels in training set.
- Action recognition in videos : CLIP better thanks to natural language supervision.
- Specialized / Complex / Abstract : requires some examples.

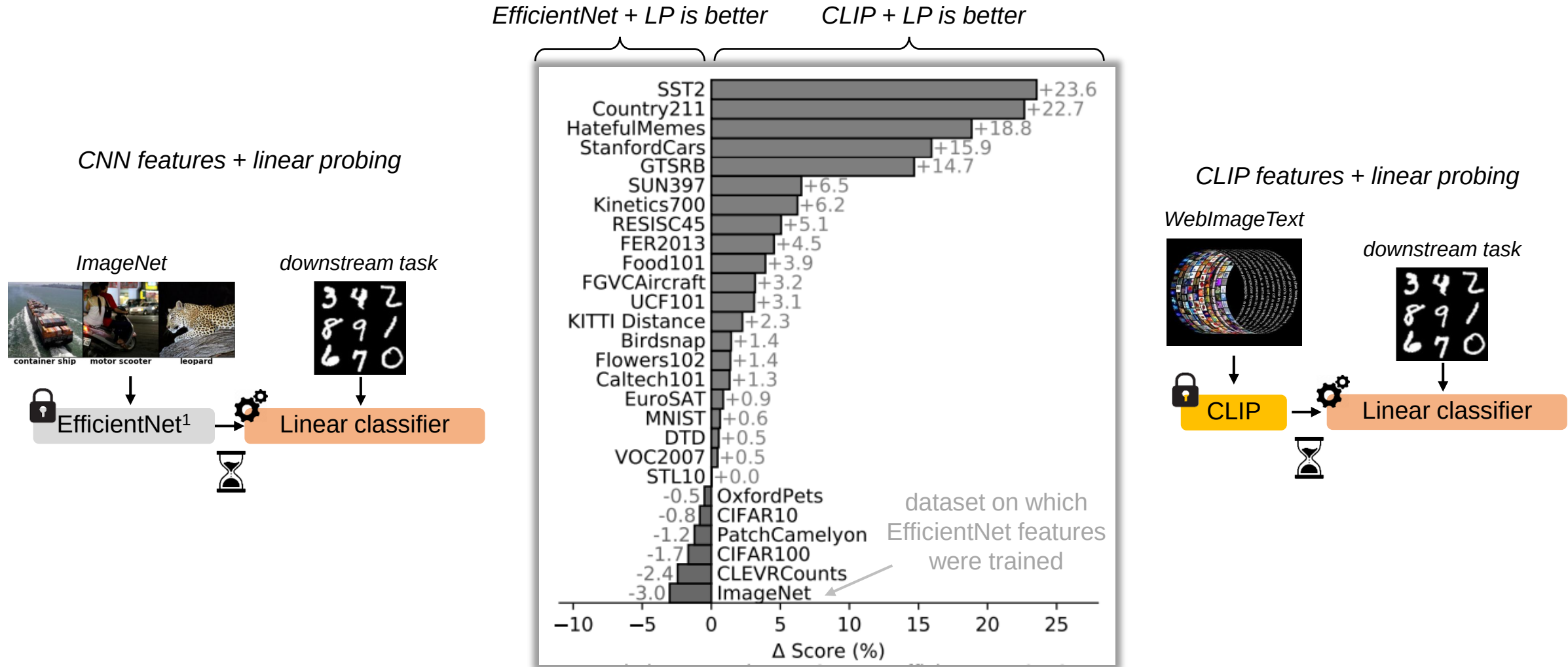


Zero-shot CLIP



We find that CLIP, similar to the GPT family, learns to **perform a wide set of tasks** during pre-training.

CLIP : Performance (linear probing)



[1] EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks, Tan (Google) (2019)

CLIP : Generalization

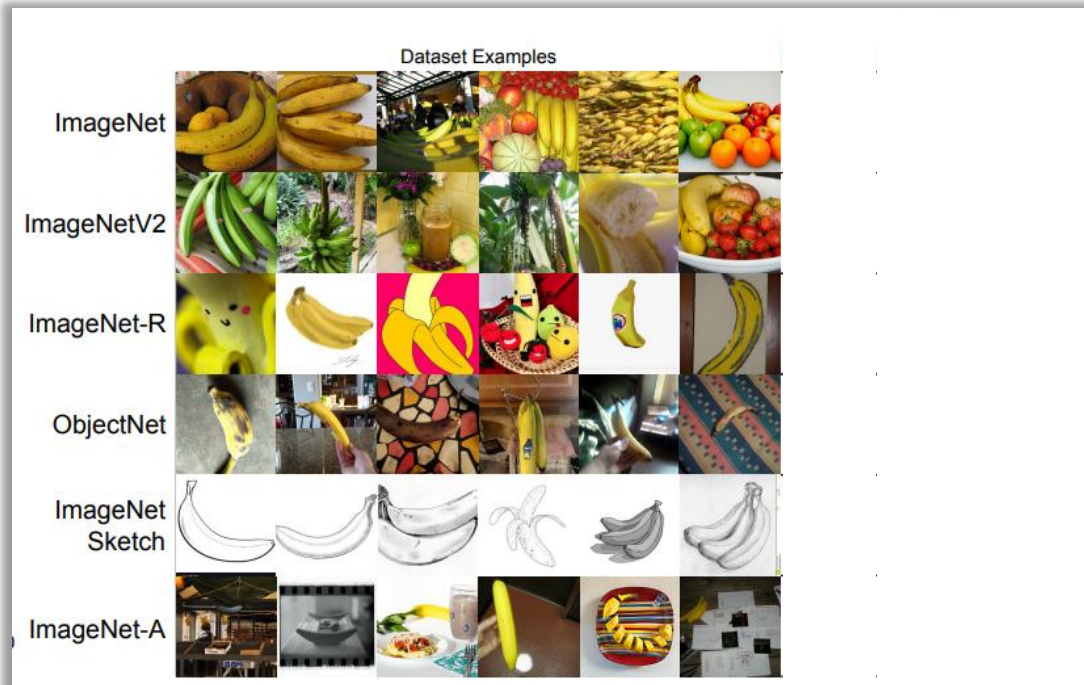
[1] Do Better ImageNet Models Transfer Better?, Kornblith (Google) (2018)

[2] Do ImageNet Classifiers Generalize to ImageNet? (2019)

Good performance on benchmark (e.g. IN) does not guarantee good performance on other distributions.^{1,2}

- **Distribution shift**

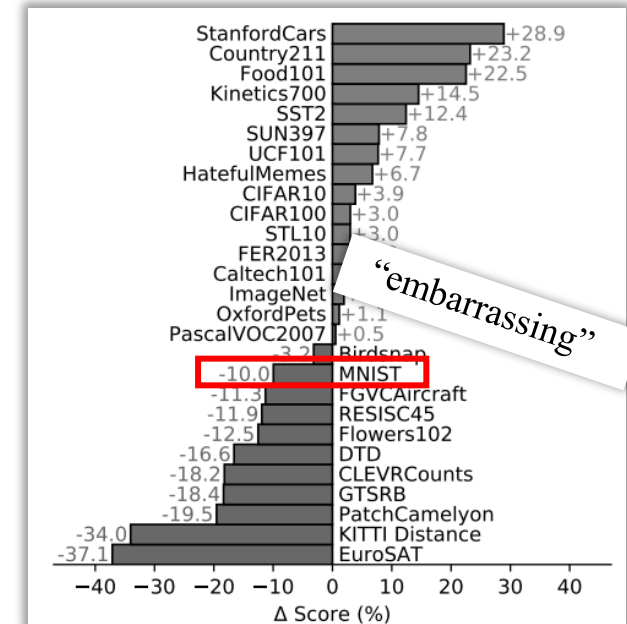
Intuitively, a zero-shot model should not be able to exploit **spurious correlations** (...) that hold only on a specific distribution, since it is **not trained on that distribution**.



The recent shift towards **large-scale task** and **dataset agnostic pre-training** combined with (...) benchmarking on **broad evaluation** suites (...) promotes the development of **more robust systems** and provides a more accurate assessment of performance.

- **Out-of-distribution generalization**

(zero-shot)



CLIP does **little** to address the underlying problem of **brittle generalization** of DL models. Instead CLIP (...) hopes that by training on such a large and varied dataset that **all data will be effectively in-distribution**. This is a naive assumption that, as MNIST demonstrates, is **easy to violate**.